UNIVERSITY DATABASE
MANAGEMENT SYSTEM
Academic Database Design & SQL Data Manipulation Project

## 1 Purpose and Requirements of the Experiment

- Database environment: SQL Server or MySQL or SQLite,etc. The database should be created, managed, and manipulated with SQL statements.
- This experiment is a design type experiment.

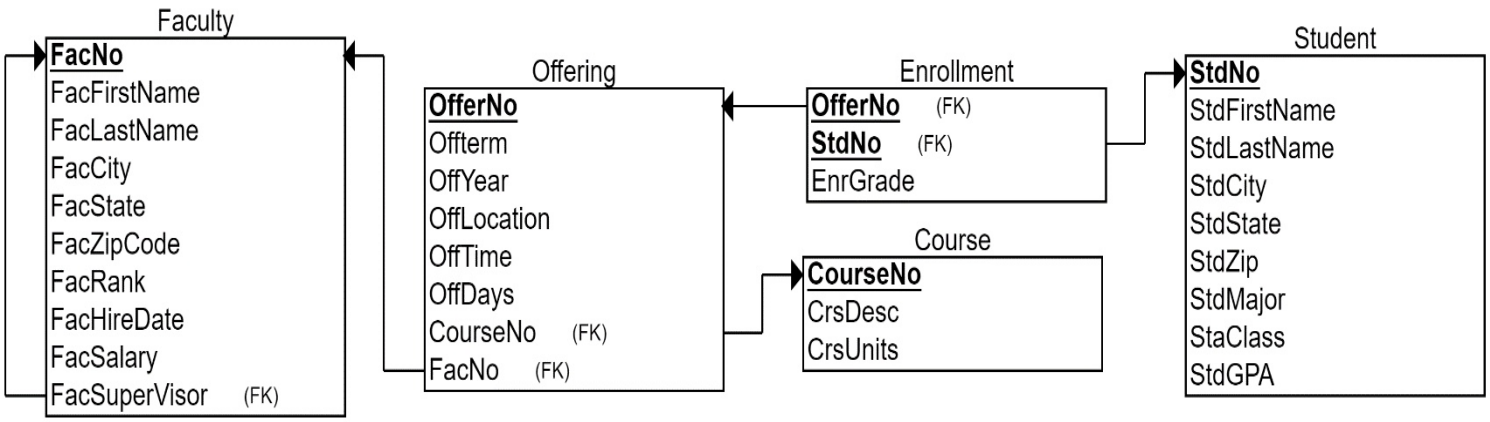
## 2 Experimental Environment

Environment: Win7 of later, SQL Server or MySQL or SQLite, etc.

## 3 The tasks of the Experiment

1) Create a database named **University.**he University database has several tables, including **student** table,**faculty** table, **course** table, **offering** table and **enrollment** table. The schema is listed below. You should create these tables using SQL statement.

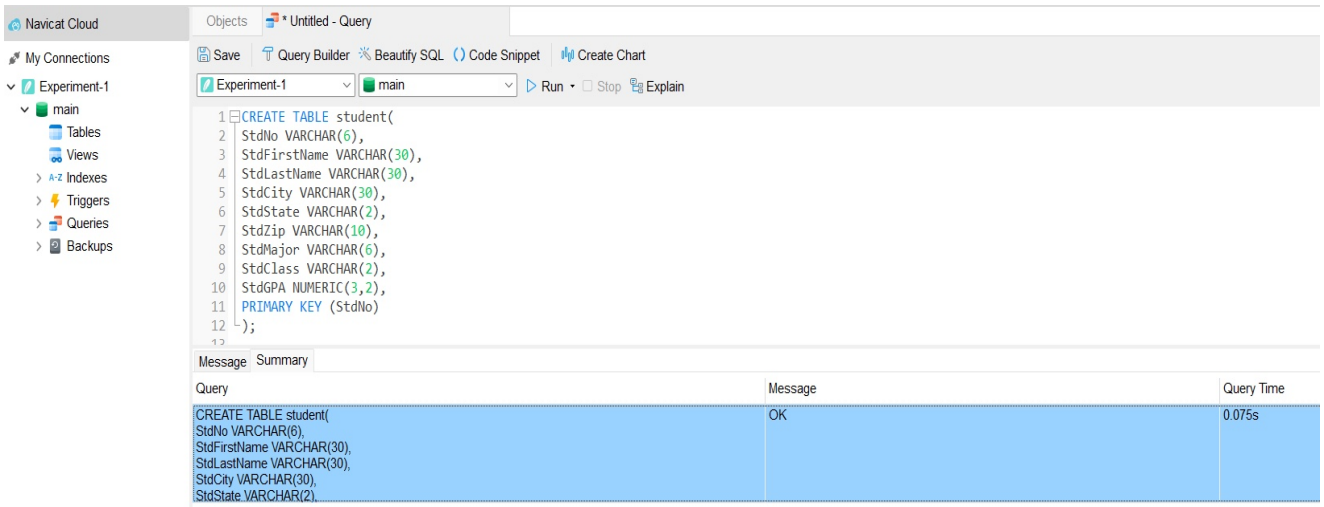
2)




The relational schema of University is:
The structure of the **student** table:

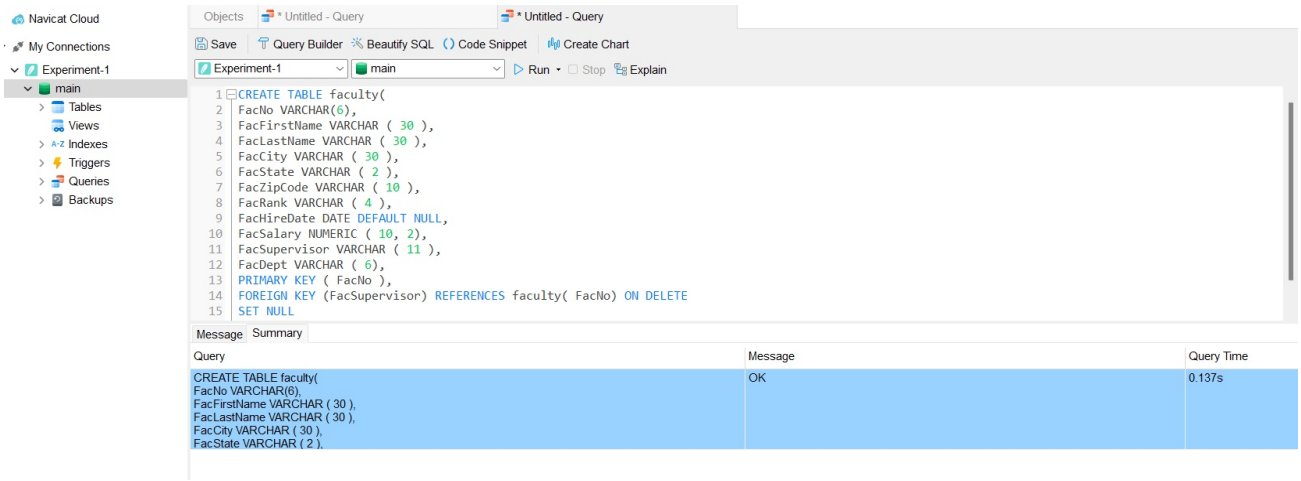
| Attributes | Data types | Length | Memo |
|---|---|---|---|

| StdNo | varchar | 6 | Primary key |
|---|---|---|---|
| StdFirstName | varchar | 30 | |
| StdLastName | varchar | 30 | |
| StdCity | varchar | 30 | |
| StdState | varchar | 2 | |
| StdZip | varchar | 10 | |
| StdMajor | varchar | 6 | |
| StdClass | varchar | 2 | |
| StdGPA | numeric | 3,2 | |



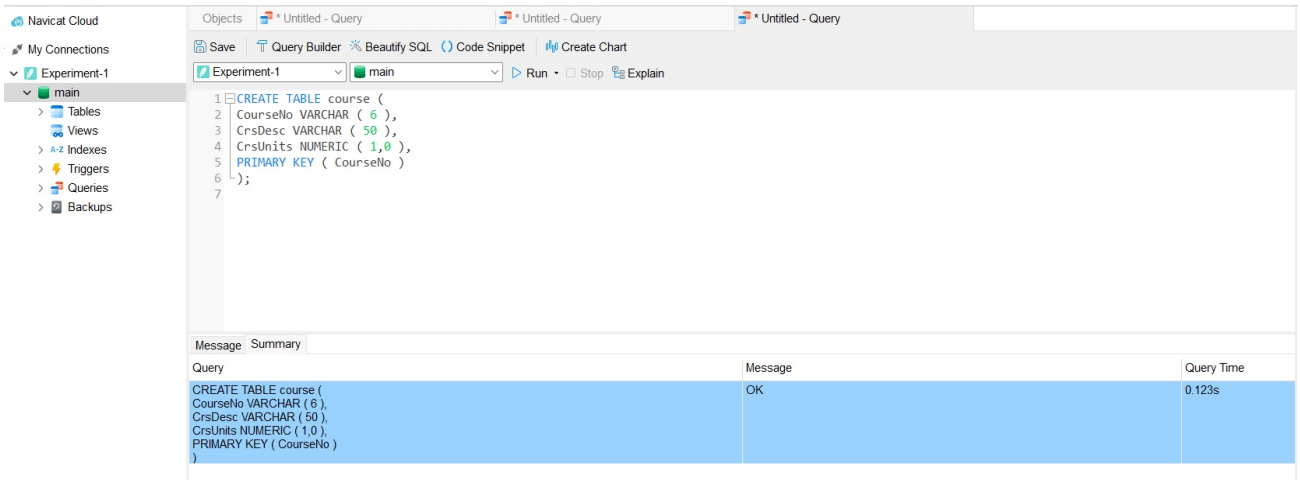

The structure of the **faculty** table:

| Attributes | Data types | Length | Memo |
|---|---|---|---|
| FacNo | varchar | 6 | Primary key |
| FacFirstName | varchar | 30 | |
| FacLastName | varchar | 30 | |
| FacCity | varchar | 30 | |
| FacState | varchar | 2 | |
| FacZipCode | varchar | 10 | |
| FacRank | varchar | 4 | |
| FacHireDate | date | | |
| FacSalary | numeric | 10,2 | |
| FacSuperVisor | varchar | 11 | |
| FacDept | varchar | 6 | |




The structure of the **course** table:

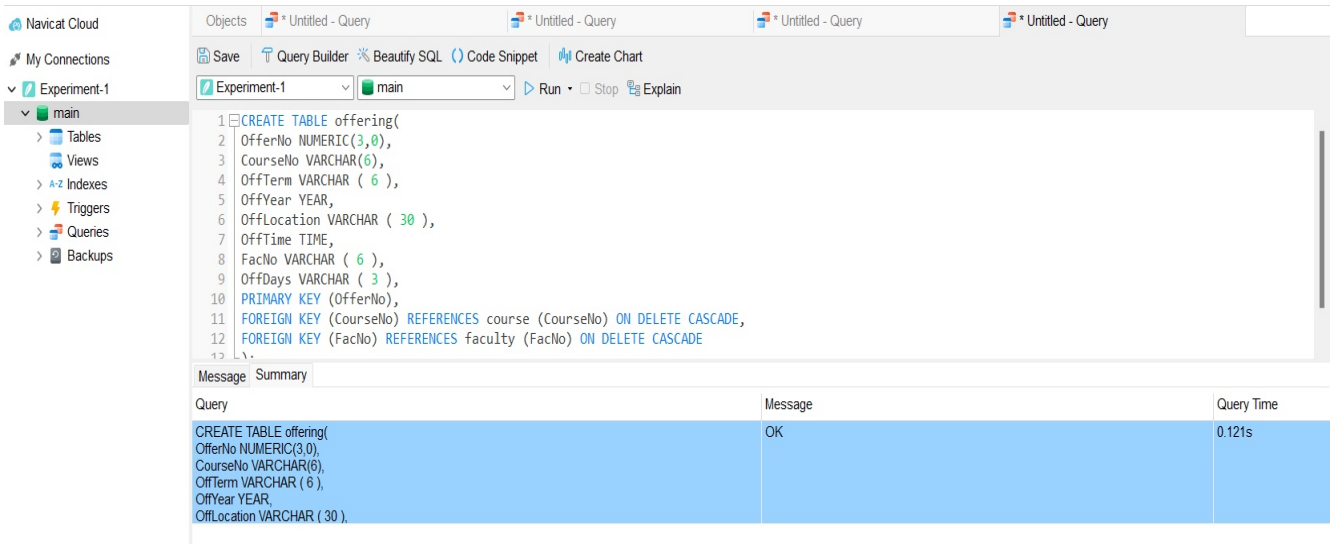
| Attributes | Data types | Length | Memo |
|---|---|---|---|
| CourseNo | varchar | 6 | Primary key |
| CrsDesc | varchar | 50 | |
| CrsUnits | numeric | 1,0 | |




The structure of the **offering** table:

| Attributes | Data types | Length | Memo |
|---|---|---|---|
| OfferNo | numeric | 3,0 | Primary key |

| | | | |
|---|---|---|---|
| CourseNo | varchar | 6 | Foreign key with the course table |
| OffTerm | varchar | 6 | |
| OffYear | year | | |
| OffLocation | varchar | 30 | |
| OffTime | time | | |
| FacNo | varchar | 6 | Foreign key with the faculty table |
| OffDays | varchar | 3 | |




The structure of the **enrollment** table:

| Attributes | Data types | Length | Memo |
|---|---|---|---|
| OfferNo | numeric | 3,0 | Primary key and foreign key with the offering table |
| StdNo | varchar | 6 | Primary key and foreign key with the student table |
| EnrGrade | varchar | 2 | |




2). use the 'insert' statement to input the following data.

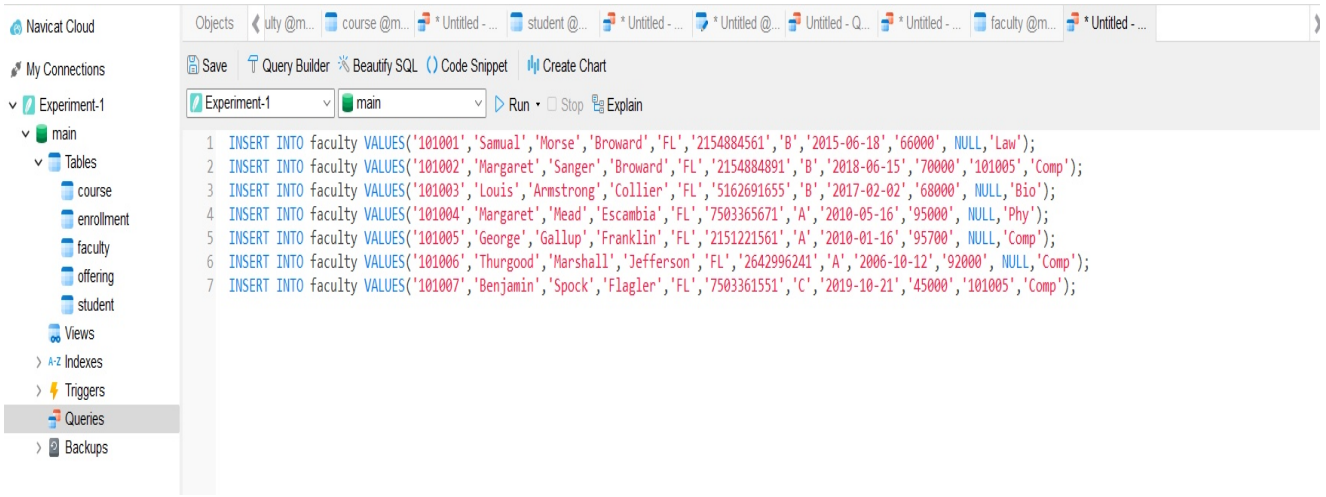
### (1) Student:

| StdNo | StdFirstName | StdLastName | StdCity | StdState | StdZip | StdMajor | StdClass | StdGPA |
|---|---|---|---|---|---|---|---|---|
| 171001 | Samuel | Morse | Broward | FL | 2154884561 | LAW | 1 | 2.80 |
| 171002 | Margaret | Sanger | Broward | FL | 2154884891 | SE | 2 | 3.20 |
| 171003 | Joseph | Smith | Collier | FL | 5162691615 | SE | 1 | 3.13 |
| 171004 | Horace | Mann | Escambia | FL | 7503365641 | CS | 1 | 1.90 |
| 171005 | Louis | Sullivan | Franklin | FL | 2151221561 | SE | 2 | 2.40 |
| 171006 | William | Faulkner | Jefferson | FL | 2642998941 | CS | 1 | 3.62 |
| 171007 | Samuel | Gompers | Flagler | FL | 7503361651 | SE | 1 | 2.64 |
| 171008 | William | James | Allentown | PA | 5283086854 | BIO | 1 | 3.50 |
| 171009 | Jane | Addams | Altoona | PA | 7503365456 | PHY | 2 | 3.20 |
| 171010 | Elvis | Presley | Harrisburg | PA | 9990395641 | PHY | 1 | 2.89 |



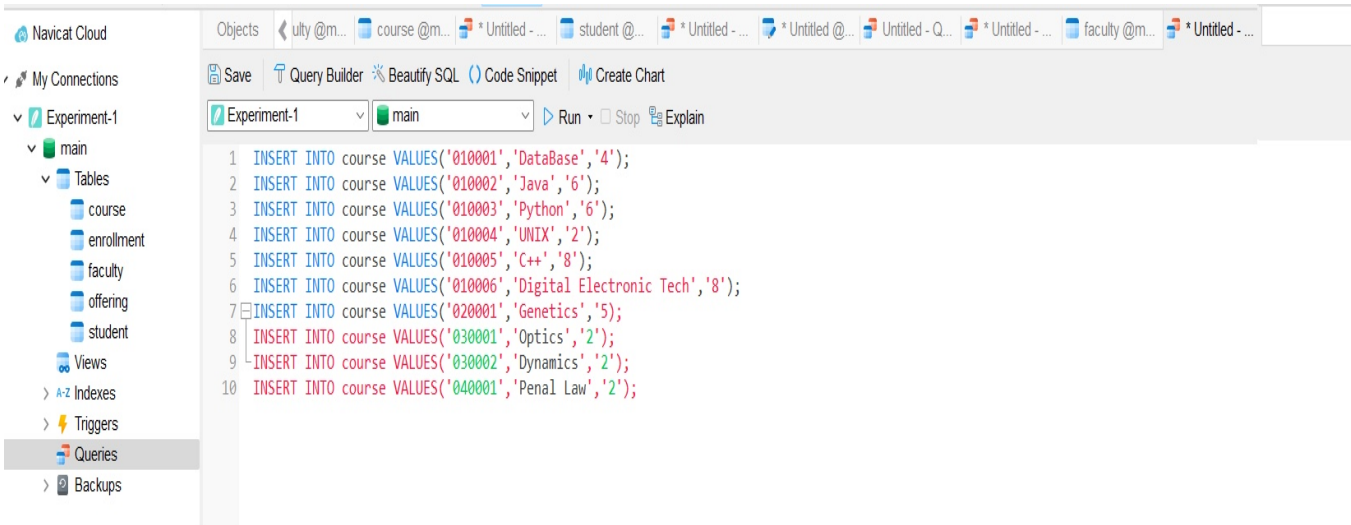

## (2) Faculty:

| FacNo | FacFirstName | FacLastName | FacCity | FacState | FacZipCode | FacRank | FacHireDate | FacSalary | FacSuperVisor | FacDept |
|---|---|---|---|---|---|---|---|---|---|---|
| 101001 | Samuel | Morse | Broward | FL | 2154884561 | B | 2015-06-18 | 66000.00 | null | Law |
| 101002 | Margaret | Sanger | Broward | FL | 2154884891 | B | 2018-06-15 | 70000.00 | 101005 | Comp |
| 101003 | Louis | Armstrong | Collier | FL | 5162691655 | B | 2017-02-02 | 68000.00 | null | Bio |
| 101004 | Margaret | Mead | Escambia | FL | 7503365671 | A | 2010-05-16 | 95000.00 | null | Phy |
| 101005 | George | Gallup | Franklin | FL | 2151221561 | A | 2005-01-16 | 95700.00 | null | Comp |
| 101006 | Thurgood | Marshall | Jefferson | FL | 2642996241 | A | 2006-10-12 | 92000.00 | null | Comp |
| 101007 | Benjamin | Spock | Flagler | FL | 7503361551 | C | 2019-10-21 | 45000.00 | 101005 | Comp |



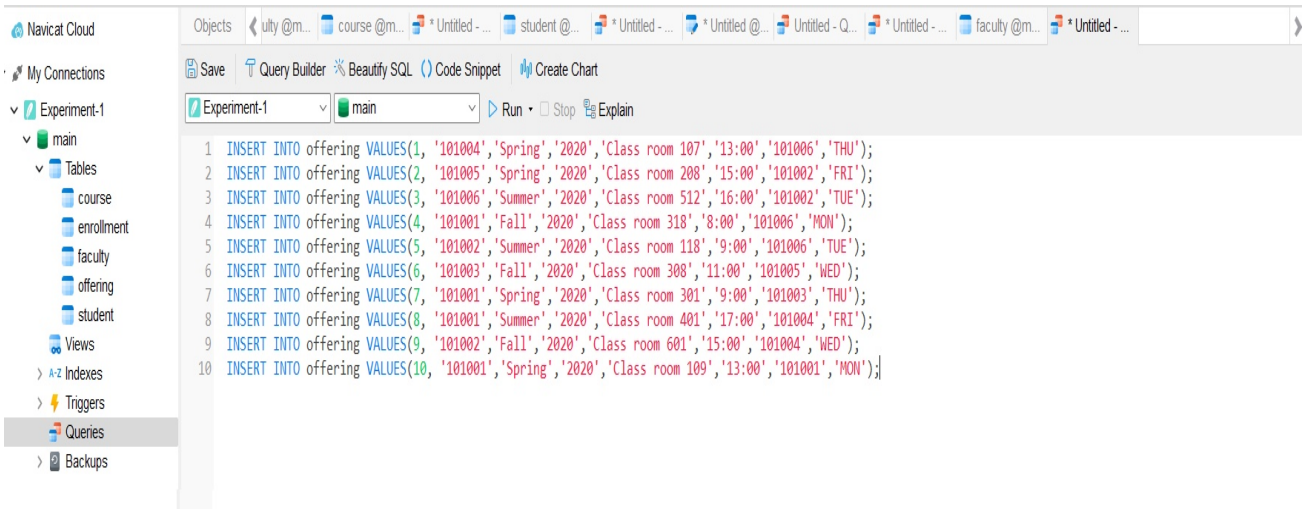

## (3) Course:

| CourseNo | CrsDesc | CrsUnits |
|---|---|---|
| 010001 | DataBase | 4 |
| 010002 | Java | 6 |
| 010003 | Python | 6 |
| 010004 | UNIX | 2 |
| 010005 | C++ | 8 |
| 010006 | Digital Electronic Tech | 8 |
| 020001 | Genetics | 5 |
| 030001 | Optics | 2 |
| 030002 | Dynamics | 2 |
| 040001 | Penal Law | 2 |



## (4) Offering:

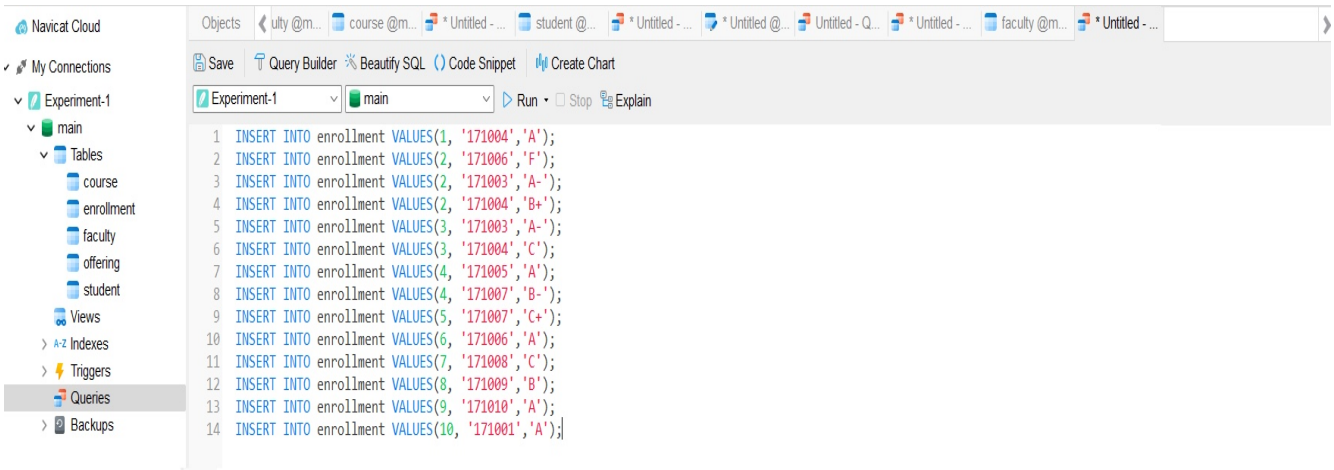
| OfferNo | CourseNo | OffTerm | OffYear | OffLocation | OffTime | FacNo | OffDays |
|---|---|---|---|---|---|---|---|
| 1 | 010004 | Spring | 2020 | Class room 107 | 13:00:00 | 101006 | THU |
| 2 | 010005 | Fall | 2020 | Class room 208 | 15:00:00 | 101002 | FRI |
| 3 | 010006 | Summer | 2020 | Class room 512 | 16:40:00 | 101002 | TUE |
| 4 | 010001 | Fall | 2020 | Class room 318 | 08:00:00 | 101006 | MON |
| 5 | 010002 | Summer | 2020 | Class room 118 | 09:00:00 | 101006 | TUE |
| 6 | 010003 | Fall | 2020 | Class room 308 | 11:00:00 | 101005 | WED |
| 7 | 020001 | Spring | 2020 | Class room 301 | 09:00:00 | 101003 | THU |
| 8 | 030001 | Summer | 2020 | Class room 401 | 17:00:00 | 101004 | FRI |
| 9 | 030002 | Fall | 2020 | Class room 601 | 15:00:00 | 101004 | WED |
| 10 | 040001 | Spring | 2020 | Class room 109 | 13:00:00 | 101001 | MON |




## (5) Enrollment

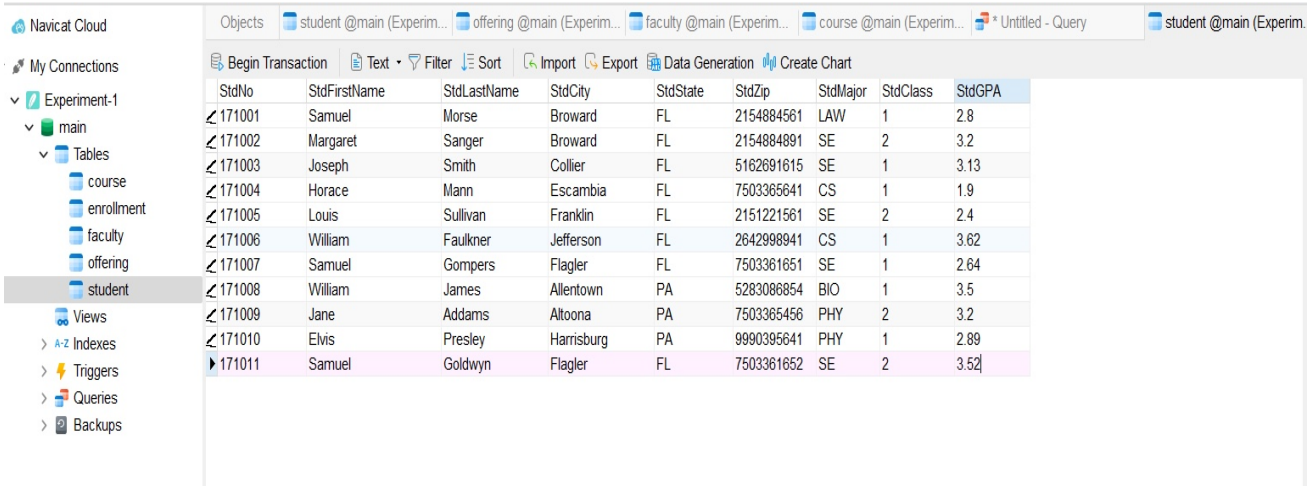
| OfferNo | StdNo | EnrGrade |
|---|---|---|
| 1 | 171002 | A |

| | | |
|---|---|---|
| 1 | 171006 | F |
| 2 | 171003 | A- |
| 2 | 171004 | B+ |
| 3 | 171003 | A- |
| 3 | 171004 | C |
| 4 | 171005 | A |
| 4 | 171007 | B- |
| 5 | 171007 | C+ |
| 6 | 171006 | A |
| 7 | 171008 | C |
| 8 | 171009 | B |
| 9 | 171010 | A |
| 10 | 171001 | A |



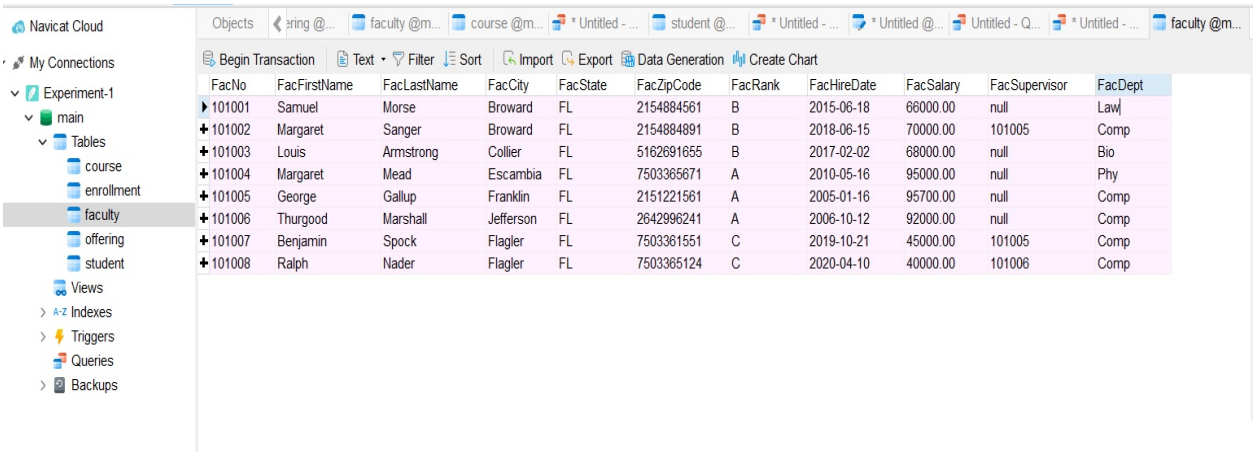

3).Use the 'insert' statement to finish the following questions and write out the results of the operation.

(1)Add a record to the student table with StdNo is 171011, StdFirstName is Samuel, StdLastName is Goldwyn, StdCity is Flagler, StdState is FL, StdZip is 7503361652, StdMajor is SE, StdClass is 2 and StdGPA is 3.52.




```
INSERT INTO student VALUES ('171011', 'Samuel', 'Goldwyn','Flagler', 'FL', '7503361652', 'SE','2','3.52');
```

(2)Add a record to the faculty table with FacNo is 101008, FacFirstName is Ralph, FacLastName is Nader, FacCity is Flagler, FacState is FL, FacZip is 7503365124, FacRank is C, FacHireDate is 2020-4-10, FacSalary is 40000, FacSuperVisor is 101006 and FacDept is Comp.




```
INSERT INTO faculty VALUES ('101008', 'Ralph', 'Nader', 'Flagler', 'FL', '7503365124', 'C', '2020-04-10', '40000', '101006', 'Comp');
```

## 4. Requirements of the Experimental Report

1). Test every SQL statement with the computer.
2). Analysis of experimental results.
3). Your harvest and experience.

REQUIREMENTS ARE FILLED ☑